



BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Real Analysis II (MAT321)

Sequential Compactness

Finite Intersection Property, Bolzano-Weierstrass Property

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CONDUCTED BY

Partho Sutra Dhor

Lecturer, BRAC University, Dhaka-1212

✉ partho.dhor@bracu.ac.bd | ✉ parthosutradhor@gmail.com

For updates subscribe on  [@ParthoSutraDhor](#)

Finite Intersection Property (FIP)

Finite Intersection Property

A family of sets $\{A_\alpha\}_{\alpha \in I}$ is said to have the *finite intersection property* if for every finite subset $J \subseteq I$, we have

$$\bigcap_{\alpha \in J} A_\alpha \neq \emptyset.$$

❓ Example of FIP

Consider the family of sets $\{A_n\}_{n=1}^{\infty}$ where $A_n = \left[-\frac{1}{n}, \frac{1}{n}\right]$. Show that this family has the finite intersection property, but the intersection of all sets in the family is empty.

② Example of FIP

Consider the family of sets $\{A_n\}_{n=1}^{\infty}$ where $A_n = [n, \infty)$. Show that this family has the finite intersection property, but the intersection of all sets in the family is empty.

Compactness and Finite-Intersection Property

💡 Compactness and FIP

A subset K of a metric space is compact if and only if for every family of closed subsets $\{F_\alpha\}_{\alpha \in I}$ of K with the finite intersection property, we have

$$\bigcap_{\alpha \in I} F_\alpha \neq \emptyset.$$

Subsequences and Limit Points

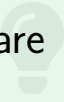
Subsequence

A *subsequence* of a sequence $\{x_n\}$ is a sequence of the form $\{x_{n_k}\}$ where $\{n_k\}$ is a strictly increasing sequence of natural numbers.

Limit Point

A point x is a *limit point* of a set A if every neighborhood of x contains at least one point of A different from x itself.

 In metric spaces, One Point imply infinitely many points

If x is a limit point of a infinite set A in a metric space, then there are infinitely many points of A in every neighborhood of x . 

Sequential Compactness and Bolzano-Weierstrass Property

Sequential Compactness

A subset K of a metric space is said to be *sequentially compact* if every sequence in K has a convergent subsequence whose limit is in K .

Bolzano-Weierstrass Property

A subset K of a metric space is said to have the *Bolzano-Weierstrass property* if every infinite subset of K has a limit point in K .

Sequential Compactness and B-W Property

A subset K of a metric space is sequentially compact if and only if it has the Bolzano-Weierstrass property. 

Compactness and Sequential Compactness

Compactness implies Sequential Compactness

Every compact metric space is sequentially compact.



Sequential Compactness implies Compactness

Every sequentially compact metric space is compact.



② Remark on Compactness and Sequential Compactness

In general, compactness and sequential compactness are not equivalent in arbitrary topological spaces. However, in metric spaces, they coincide. Provide an example of a topological space where compactness does not imply sequential compactness.

Thank You!

We'd love your questions and feedback.

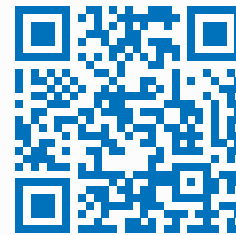
Partho Sutra Dhor

Lecturer, BRAC University, Dhaka-1212

✉ partho.dhor@bracu.ac.bd | ✉ parthosutradhor@gmail.com

 **@ParthoSutraDhor**

(Lectures, walkthroughs, and course updates)



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References

- [1] Erwin Kreyszig, *Introductory Functional Analysis with Applications*, John Wiley & Sons, 1978.
- [2] Walter Rudin, *Principles of Mathematical Analysis*, 3rd Edition, McGraw-Hill, 1976.